CS-302: Artificial Intelligence

LECTURE NO 7:

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Organization of Lecture

1. Informed Search Strategies (***Continuation of Lecture 6***)
    1. Beam Search
    2. A* (A-Star) Algorithm
    3. AO* (AND-OR-Star) Algorithm

1. Beam Search (1/4)

1) Beam Search

optimized Version of $\boxed{\text{Best First Search.}}$

↳ Heuristic Search Algorithm.

↳ Explores a Graph by expanding the most promising node in a $\boxed{\text{limited Set.}}$

↳ Reduces Memory Requirement. (Limited/Less no. of selected candidates will result in less memory requirement).

↳ Greedy Algorithm ⇒ An algorithm for solving a problem by selecting the best option available at the moment. It doesn't worry whether the current best result will bring the overall optimal result.

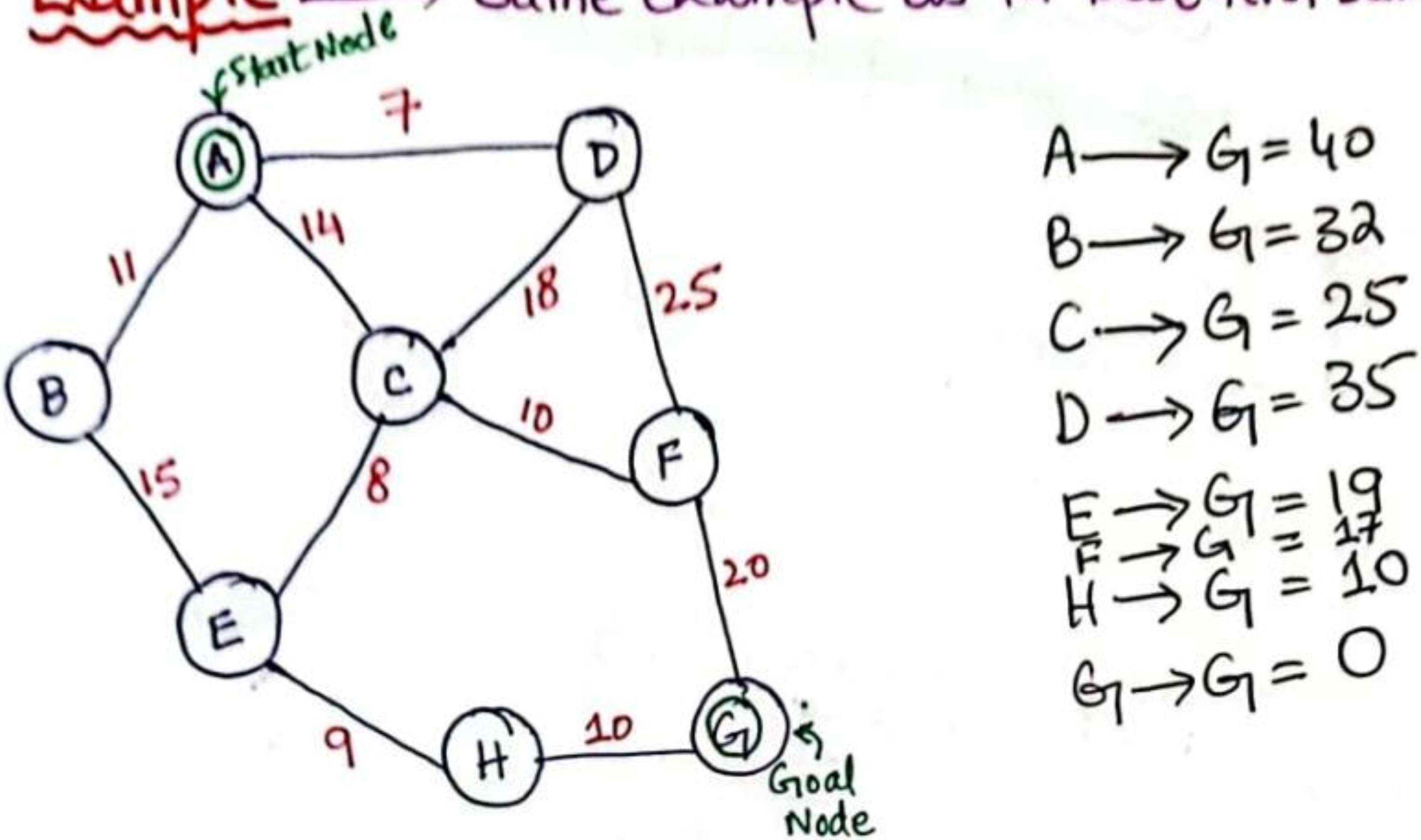
1. Beam Search (2/4)

↳ Predetermined no. of best partial solutions are kept as Candidates.

↳ Predetermined no. is also called Beam Value (β).

1. Beam Search (3/4)

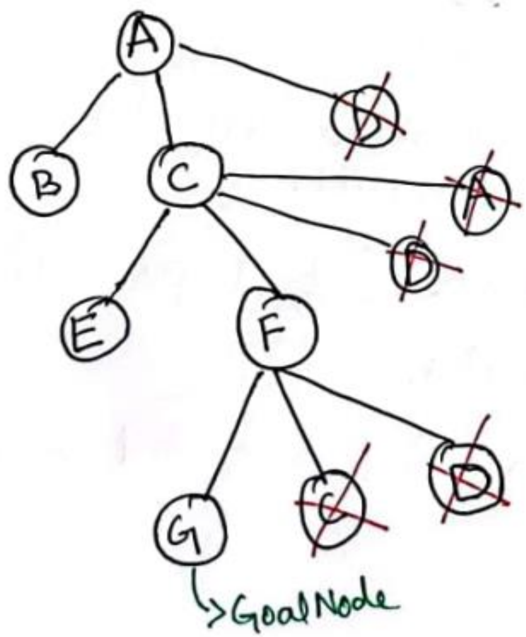
Example ⟹ Same example as in Best First Search.



1. Beam Search (4/4)

Solution

Let us suppose value of 'β' is given as 2 i.e., $\beta = 2$ (Best two nodes will be considered)



↳ We may receive the same solution as BFS but we achieve less memory requirement with the help of beam value (β).

↳ If we set $\beta = 1$, then it will become Hill Climbing Algorithm.

2. A* (A-Star) Algorithm (1/8)

2) A* Algorithm

- ↳ Uses heuristic function 'h(n)' and cost to reach the node 'n' from start state 'g(n)'.
- ↳ Finds shortest path through search space.
- ↳ Fast and optimal result

$$\underbrace{f(n)}_{\text{Total Estimated Cost}} = g(n) + \overbrace{h(n)}^{\text{heuristic value (child node)}}$$

$g(n)$ ↳ Cost to reach node 'n'

2. A* (A-Star) Algorithm (2/8)

Algorithm:

↳ i) Enter starting node in OPEN List.

ii) If OPEN List is empty return FAIL.

iii) Select node from OPEN list which has smallest value (g+h)

↳ if node = Goal, return Success.

iv) Expand node 'n' and generate all Successors.

↳ compute (g+h) for each Successor node.

v) If node 'n' is already in OPEN/CLOSED, attach to back pointer.

vi) Go to (ii).

2. A* (A-Star) Algorithm (3/8)

Advantages:-

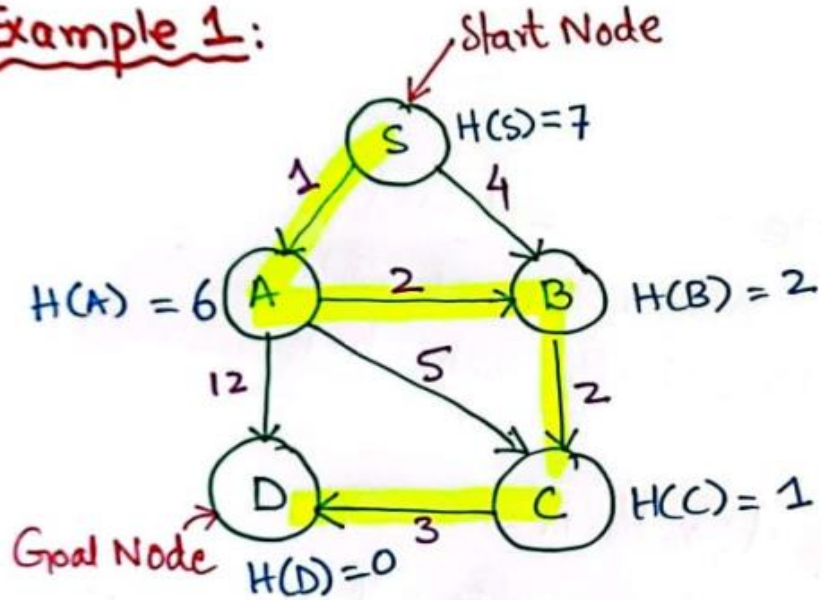
i) Best Searching algorithm
ii) Optimal and Complete
iii) Solving complex problems.

Disadvantages:-

i) Doesn't always produces shortest path.
ii) Complexity issue in its implementation.
iii) Required memory.

2. A* (A-Star) Algorithm (4/8)

Example 1:



$S \longrightarrow A = g(n) + h(n) = 1+6 = 7$

$S \longrightarrow B = g(n) + h(n) = 2+4 = 6$ ✓ → (Explored)

heuristic value of child 'C'

$S \rightarrow B \rightarrow C = (\underbrace{4}_{\text{"g" value from } S \rightarrow B} + \underbrace{2}_{\text{"g" value from } B \rightarrow C}) + \underbrace{1}_{h(n)} = 7$ ✓ → (Explored)

"g" value from S → B

"g" value from B → C.

'g' values

h value of child 'D'

$$\boxed{S \longrightarrow B \longrightarrow C \longrightarrow D = \overbrace{(4+2+3)}^{\text{'g' values}} + \overset{\text{h value of child 'D'}}{0} = 9}$$

R1

2. A* (A-Star) Algorithm (5/8)

Since **S** $\longrightarrow$ A is not explored yet. Let's explore it

$S \longrightarrow A \longrightarrow B = (1+2)+2 = 5$ ✓ Explored

$S \longrightarrow A \longrightarrow C = (1+5)+1 = 7$

$\boxed{S \longrightarrow A \longrightarrow D = (1+12)+0 = 13}$ R2.

Let's now explore $S \longrightarrow A \longrightarrow B = 5$ (Small/less estimated cost)

$S \longrightarrow A \longrightarrow B \longrightarrow C = (1+2+2)+1 = 6$ ✓ Explored

further explore $S \longrightarrow A \longrightarrow B \longrightarrow C$ i.e.,

$\boxed{S \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow D = (1+2+2+3)+0 = 8}$ R3

2. A* (A-Star) Algorithm (6/8)

Now only 1 route is left/yet to be explored i.e.,

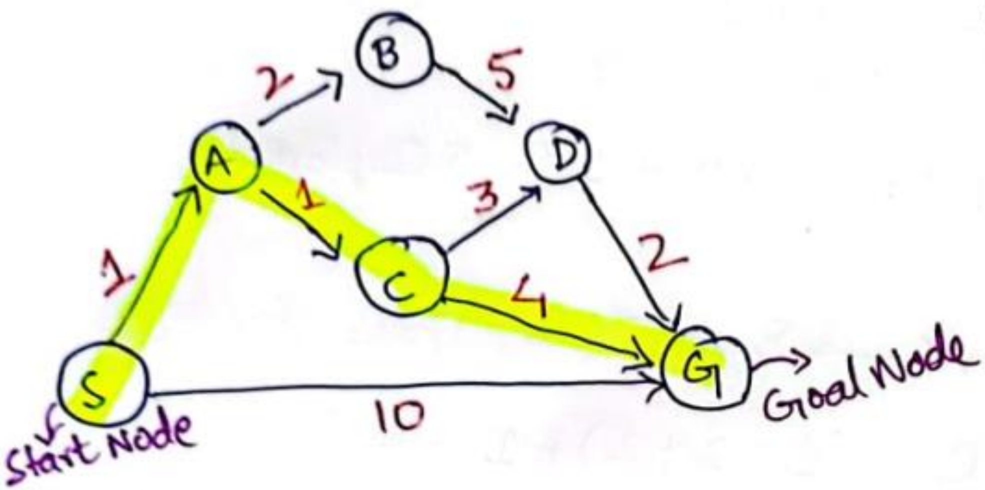
$$\boxed{S \rightarrow A \rightarrow C \rightarrow D = (1+5+3)+0 = 9} \quad R4$$

Finally we have come up with 4 routes i.e R1(9), R2(13), R3(8) & R4(9). We will select the route with minimum estimated cost i.e R3 = 8 i.e.,

$$S \rightarrow A \rightarrow B \rightarrow C \rightarrow D = (1+2+2+3)+0 = 8$$

2. A* (A-Star) Algorithm (7/8)

Example 2



State	h(n)
S	5
A	3
B	4
C	2
D	6
G	0

Solution

$S \to A = \overset{g(A)}{1} + \overset{h(A)}{3} = 4$

$\boxed{S \to G = \overset{g(G)}{10} + \overset{h(G)}{0} = 10}$ → Reached Goal Node. (Not further Explored) (R1)

$S \to A \to B = \overbrace{(1+2)}^{\text{'g' values}} + \overset{\text{'h' value}}{4} = 7$ ✓

$S \to A \to C = (1+1) + 2 = 4$ ✓

2. A* (A-Star) Algorithm (8/8)

$S \rightarrow A \rightarrow C \rightarrow D = (1+1+3)+6 = 11$ ✓

$\boxed{S \rightarrow A \rightarrow C \rightarrow G = (1+1+4)+0 = 6}$ → Reached Goal Node, Not further Explored (R2)

$S \rightarrow A \rightarrow B \rightarrow D = (1+2+5)+6 = 14$

$\boxed{S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = (1+1+3+2)+0 = 7}$ Reached Goal Node (R3)

$\boxed{S \rightarrow A \rightarrow B \rightarrow D \rightarrow G = (1+2+5+2)+0 = 10}$ Reached Goal Node, (R4)

The route with minimum cost will be selected i.e. R2

$\boxed{R2 = S \rightarrow A \rightarrow C \rightarrow G = 1+1+4 = 6}$

3. AO* (AND-OR-Star) Algorithm (1/4)

3) AO* Algorithm

AND-OR Graphs:-

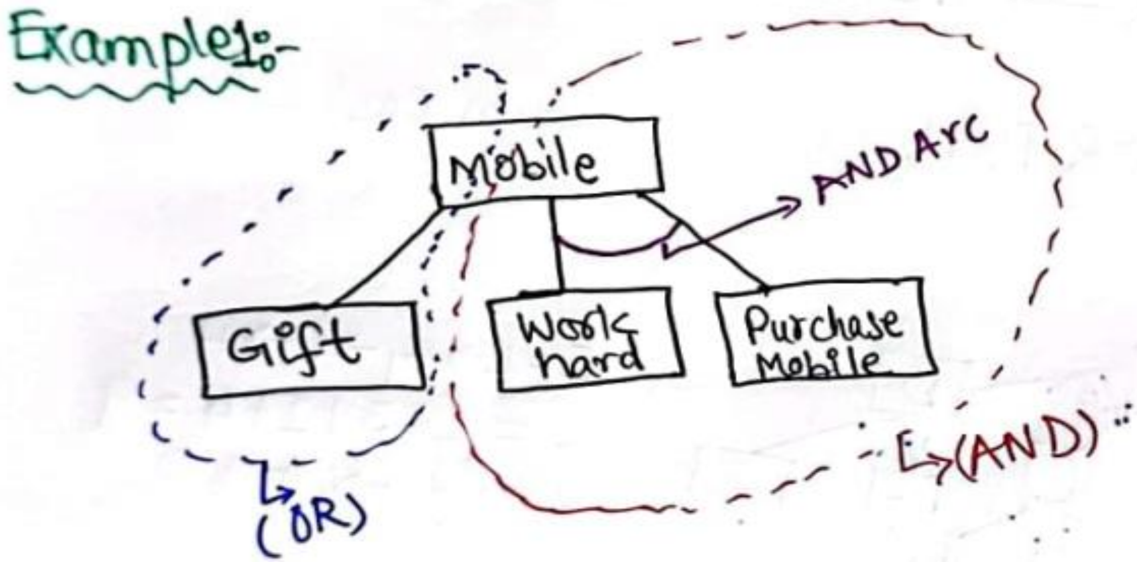
↳ Useful for representing the probelm solution that can be solved by decomposing them into Smaller set of probelms all of which must be solved.

↳ Decomposing ⟶ AND ARCS ⟶ May point to 'n' no. of Successor nodes.

↳ Algo. like Best-Fit search can be used that has the ability to handle AND Arcs.

3. AO* (AND-OR-Star) Algorithm (2/4)

Example1:-

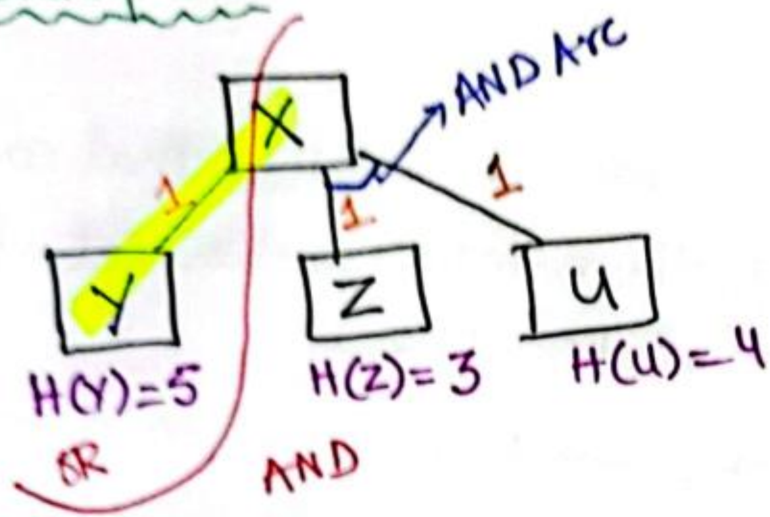


[FUTILITY: Should be chosen to correspond to a threshold value. If $\underbrace{\text{estimated cost} > \text{Futility}}_{\text{Stop Search}}$]

↳ If cost of OR part is minimum, we would choose OR part otherwise we would choose AND part.

3. AO* (AND-OR-Star) Algorithm (3/4)

Example 2:-



OR Part

$X \rightarrow Y = 1+5=6$ ✓

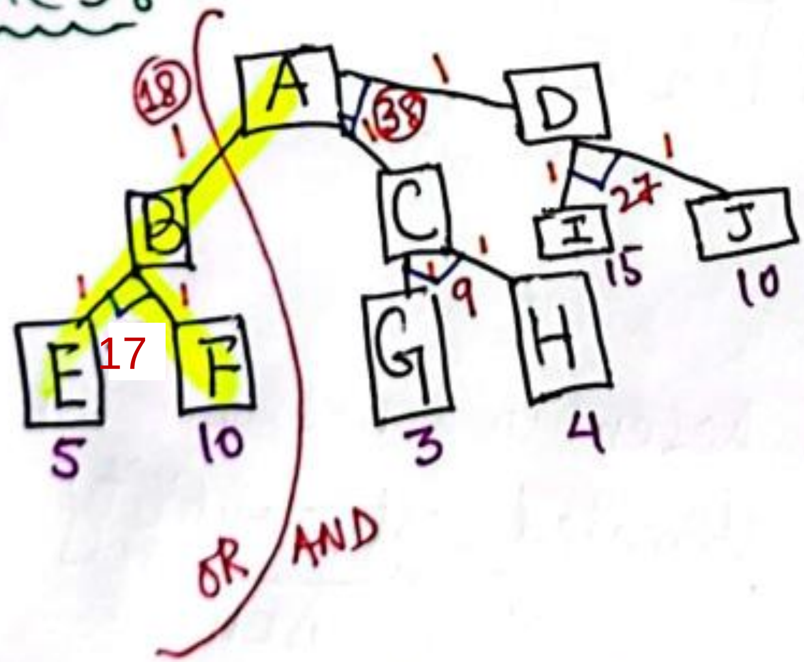
AND Part

$X \rightarrow Z \rightarrow X \rightarrow U = 1+3+1+4 = 9$

→ We will chose OR Part (with minimum cost = 6).

3. AO* (AND-OR-Star) Algorithm (4/4)

Example 3:-



↳ The OR part will be chosen (it has minimum cost i.e. 18)

References: Textbooks

1. *"Artificial Intelligence", by Elaine Rich and Kevin Knight, (2006), McGraw Hill companies Inc., Chapter 1-22 page 1-613.*
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3. *"Computational Intelligence: A logical Approach", by David Poole, Alan Mackworth, and Randy Goebel, (1998), Oxford University Press, Chapter 1-12, page 1-608.*
4. *"Artificial Intelligence: Structures and Strategies for Complex Problem Solving", by George F.Lugar, (2002), Addison-Wesley, Chapter 1-16, page 1-743.*
5. *"AI: A new Synthesis", by Nils J.Nilsson, (1998), Morgan Kaufmann Inc., Chapter 1-25, Page 1-493.*
6. *"Artificial Intelligence: Theory and Practice" by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.*
7. *Related documents from open source, mainly internet.*